each domain element appears in the operation tables[20], but the user can include any number of other discriminators. *Isofilter* "as is" handles models of size up to around 10 with ease; but the combinatorial explosion soon makes computations unfeasible for larger sizes, unless discriminators well suited to the problem are chosen. For example, *Mace4* provides 10 models of size 15 for E1518 $\not\models$ E47, of which 6 belong to different isomorphism classes. Vanilla *Isofilter* requires 10 days and processes $3 \cdot 10^{13}$ permutations in order to determine these classes, while adding the discriminators `x*x=x. (x*y)*z = x*(y*z). x*y=y. x*y=y*x.` reduces the task to 10^5 permutations, completed in 0.02s.

8. Implications for finite magmas

Many of the techniques used to determine the graph of implications E $\models$ E′ can also be used to determine the graph of finite implications E $\models_{\text{fin}}$ E′, with the notable exception of the greedy construction, which appears to be inherently infinitary in nature. On the other hand, when the magma $\mathcal{M}$ is finite, one can prove additional implications by using the fact that any function $f\colon M \to M$ which is surjective is necessarily injective, or vice versa. We could establish about 200 new implications by providing these two axioms to *Vampire* or to the *Lean* package *Duper*, though in the latter case some human rewriting of the proof was needed to formalize it in the base installation of *Lean*. A small number of additional implications could be resolved by more complicated facts about functions $f, g\colon M \to M$, such as the fact that $f = f \circ f \circ g$ implies $f = f \circ g \circ f$. We refer the reader to the blueprint for examples of such arguments, which were obtained by *ad hoc* methods.

In the end, we were able to establish 820 new implications E $\models_{\text{fin}}$ E′ for which E $\not\models$ E′; and for most other anti-implications E $\not\models$ E′, we were able to strengthen the anti-implication to E $\not\models_{\text{fin}}$ E′. However, there was (up to duality) precisely one finite implication which we could not settle, which we leave as an open problem:

Problem 8.1. *Does the law* $\mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{x} \diamond ((\mathrm{y} \diamond \mathrm{x}) \diamond \mathrm{y}))$ (E677) *imply the law* $\mathrm{x} \simeq ((\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{x}) \diamond \mathrm{x}$ (E255) *for finite magmas?*

This problem appears to be "immune" to many of our constructions, such as the linear magma construction or the magma cohomology construction; the greedy construction does show that E677 $\not\models$ E255, but the construction is inherently infinite in nature. We tentatively conjecture that E677 $\not\models_{\text{fin}}$ E255; we refer the reader to the blueprint for several partial results in this direction.

9. Spectrum of equational laws

Given an equational law En, one can ask for its spectrum, namely the set of cardinalities of its finite models.[21] The spectrum Spec(En) is a multiplicative subset of $\mathbb{Z}_{>0}$ since the direct

[20]This discriminator is entirely ineffective with models having same number of copies of each element in their tables, such as quasigroups.

[21]The infinite spectrum is uninteresting (assuming the axiom of choice). Let us show that if En is not equivalent to the singleton law E2 then it has models of all infinite cardinalities κ. The free magma on En

product of models is a model. We focus here on the most basic question, that is, which laws (of order up to 4) have spectrum equal to $\mathbb{Z}_{>0}$?

Several extensions will be described in a separate publication: determining the spectrum and not only whether it is full; the spectrum of simple magmas or (sub)directly irreducible magmas; tracking multiplicity, namely counting (or bounding asymptotically) how many finite magmas exist of each size in the spectrum; this is referred to as the *fine spectrum* in [67]. These detailed considerations reveal profound differences in how much an equational law constrains the magma operation, and may help organize the implication graph into different families.

An ATP run shows that 1558 laws of order up to 4 have no model of size 2 and 62 have a model of size 2 but none of size 3. Pushing the search to higher model sizes does not resolve the question for any of the remaining 3074 laws. As we explain next, all of these laws actually have full spectrum.[22]

Our main tool by far to show that a law has full spectrum is to consider the carrier set $\mathbb{Z}/n\mathbb{Z}$, with a linear operation $x \diamond y = ax + by$ with $a, b \in \{-1, 0, 1\}$. If the law holds for some choice of a, b then the law has full spectrum.

- For $(a, b) = (0, 0)$ the operation is the constant operation, which is a model of E1 and any law whose sides both have positive order. Equivalently, these laws are consequences of the constant law E46.
- For $(a, b) = (1, 0)$ the operation is a projection, which is a model of any law whose sides start with the same first variable, equivalently the consequences of E4. (The choice $(a, b) = (-1, 0)$ is a model of fewer laws hence is not useful.) Likewise $(a, b) = (0, 1)$ shows that laws whose sides end with the same last variable (consequences of E5) have full spectrum.
- For $(a, b) = (1, -1)$ the operation is abelian group subtraction, characterized by Tarski's axiom E543, which shows that any law implied by E543 has full spectrum. Likewise, backwards subtraction $(a, b) = (-1, 1)$ provides models for E1090 (equivalent to the dual of E543) and its consequences.
- The operation for $(a, b) = (1, 1)$ is abelian group addition, which is useless as any law it satisfies for all n is also satisfied by the constant operation $(a, b) = (0, 0)$.
- Finally, the operation $(a, b) = (-1, -1)$ is a model of some more laws, such as the semi-symmetric quasigroup law E14 and totally symmetric quasigroup law E492.

These considerations account for 3068 laws, and there remain three dual pairs of laws to treat. This is done through ad hoc models: a piecewise linear model for E1682 and its dual, and models whose operation table is mostly constant for the remaining laws E1482, E1523, and their duals.

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with κ generators is such a model. Indeed, the generators are distinct in this magma (otherwise En would imply E2) so its cardinal μ is at least κ. Conversely, $\mu \leq \sum_T \kappa^{|T|}$ where the range sums over finite binary trees, and this sum is bounded above by $\aleph_0 \cdot \kappa = \kappa$.

[22] Further investigations show that the lowest-numbered law with models of sizes 2 and 3 but not full spectrum is E80887, namely $\mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{y} \diamond (\mathrm{y} \diamond (((\mathrm{y} \diamond \mathrm{y}) \diamond \mathrm{x}) \diamond \mathrm{y})))$, of order 6.

Since a law implied by a full spectrum law has full spectrum itself, the implication graph reduces significantly the number of laws for which it is useful to formalize the full spectrum property. Accounting for duality and implications, we found it sufficient to formalize the proof that 32 laws have no magma of size 2 or none of size 3, and the explicit construction of magmas of all finite sizes for the 7 laws E4, E41, E492, E543, E1482, E1523, and E1682. In conclusion, we prove that 3074 laws (65%) have full spectrum $\mathrm{Spec}(\mathrm{E}n) = \mathbb{Z}_{>0}$ and 1620 (35%) do not (including 1496 laws equivalent to E2). These percentages remain roughly stable at higher orders, with 60% of laws of order up to 9 having full spectrum, as will be reported elsewhere.

10. Higman–Neumann laws

10.1. **Describing groups as magmas.** The ETP is focused exclusively on magmas, which only feature a single (binary) operation. Many mathematical structures traditionally defined using several operations can nevertheless be fully described as magmas with a well-chosen combined operation, from which the whole structure can be reconstructed. The first example is how Boolean algebras defined in terms of three operations $(\wedge, \vee, \neg)$ were equivalently described in 1913 in terms of the Sheffer stroke $x \diamond y \coloneqq \neg(x \wedge y)$ [61]. Once such a single operation is found, a separate endeavor is to determine which laws it must satisfy to get the desired structure, and, in favorable cases find a single law that encapsulates the whole structure, or even find all equivalent laws of minimum order. The earliest such example is Tarski's description of abelian groups in terms of subtraction $x \diamond y \coloneqq x + (-y)$, subject to a single axiom $\mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{z} \diamond (\mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{z})))$ (E543), found in 1938 [66]. It then took three decades [29, 62, 57] to sort out the full equivalence class of E543 among laws of order 4. For Boolean algebras, a minimum-order single-law description was only found in [47], nine decades after Sheffer's work.

We plan to report elsewhere on other examples such as modules over Eisenstein integers $\mathbb{Z}[\omega_3]$ or Gaussian integers $\mathbb{Z}[\omega_4]$, with ω_k a primitive kth root of unity, which can be described by the operation $x \diamond y \coloneqq x + \omega_k y$ subject to the order-6 laws E85914 and E86082, respectively.

Here, we describe the case of groups. The binary operation $*$, unary operation $(\cdot)^{-1}$, and nullary operation e (identity element) can be repackaged into a single division operation $x \diamond y \coloneqq x * y^{-1}$, from which the original operations are easily reconstructed: for instance $x * y = x \diamond ((y \diamond y) \diamond y)$. A group equipped with division, called a Ward quasigroup, is a magma $(G, \diamond)$ satisfying the unipotence law $\mathrm{x} \diamond \mathrm{x} \simeq \mathrm{y} \diamond \mathrm{y}$ (E40), the right-unit squares law $\mathrm{x} \simeq \mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{y})$ (E11), and a version of the associativity law dubbed the half-group law, $\mathrm{x} \diamond \mathrm{y} \simeq (\mathrm{x} \diamond \mathrm{z}) \diamond (\mathrm{y} \diamond \mathrm{z})$ (E3737), from which group axioms are easily derived. These three laws are equivalent to a single law $\mathrm{E}_{\mathrm{HN}} \coloneqq$ E42323216 of order 8, found by Higman and Neumann [29],

$$\mathrm{E}_{\mathrm{HN}} \colon \mathrm{x} \simeq \mathrm{y} \diamond \Big(\big(((\mathrm{y} \diamond \mathrm{y}) \diamond \mathrm{x}) \diamond \mathrm{z}\big) \diamond \big(((\mathrm{y} \diamond \mathrm{y}) \diamond \mathrm{y}) \diamond \mathrm{z}\big)\Big).$$

McCune found two more laws equivalent to this one and of the same order [48], E42302852 and E147976245. A natural question is to find all characterizations of Ward quasigroups (groups equipped with division) with minimum order. Throughout our exploration, we used